

Greening while Governing the Economy: Central Banks Matter to Decarbonization*

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Abstract

Why do some countries translate international climate commitments into emissions reductions while others do not? I test whether the answer lies in central bank regulatory and governance powers. Membership in the Network for Greening the Financial System (NGFS)—a voluntary international network—is used as a measure of climate-oriented financial governance across 145 countries from 2013 to 2023. Two-way fixed-effects models and doubly robust difference-in-differences estimates show that NGFS membership reduces within-country emissions by 4 to 9 percent, depending on the modeling technique employed. The effect is small where central banks have low independence from political authority and nearly doubles where independence is high. Mediation analysis is consistent with independence functioning as a precondition for the network’s effectiveness rather than a consequence of membership. International regulatory networks matter for decarbonization, but only where domestic institutions, statutes, and regulations allow central banks to act.

*Acknowledgements.

Why do some countries translate international climate commitments into emissions reductions while others do not? One answer relates to the role central banks play in directing investments to low-carbon markets (1–4). Central banks regulate how trillions of dollars flow from banks and investment funds into the economy, governing which sectors receive capital and according to what rules, statutes, and norms (5–8). Yet, whether and how financial supervisors facilitate or hinder the low-carbon transition remains an ongoing discussion (1–4, 9, 10). I illuminate this discussion by estimating the extent to which nations internationally committed to “green” their central banks emit less greenhouse gases (GHG) than nations with conventional central banks.

Addressing the question above by focusing on financial supervisors has pressing policy implications. Confronting the climate crisis requires decarbonizing not only factories, value chains, and power grids but also the financial systems that fund them. A challenge for the empirical analysis of central banks’ influence on decarbonizing the global economy is that no consistent cross-national marker has distinguished countries with “green” central banks from nations with conventional ones. This changed in 2017, when the Network of Central Banks and Supervisors for Greening the Financial System (NGFS) was established (10, 11).

The NGFS offers the first public registry of financial regulators formally and internationally committed to integrating climate risks into macroeconomic supervision (11). Using NGFS membership as a treatment indicator, I compare GHG emission trajectories across 145 countries from 2013 to 2023 ($N = 1,595$). To account for the fact that countries joined the NGFS in different years—i.e., staggered treatment timing—I employ the Callaway and Sant’Anna (12) doubly robust estimator, which isolates cohort-specific treatment effects and is robust to the heterogeneous treatment timing that standard two-way fixed effects models fail to handle without bias. Difference-in-differences models with two-way fixed effects reveal that, within countries, joining the NGFS is associated with a subsequent deceleration in emission growth. This reduction ranges from approximately 4% to 9% lower emissions relative to what within-country trends would otherwise predict, after controlling for economic, ecological, and political factors.

The effect above grows with central bank independence (13, 14). NGFS membership carries little measurable impact where central banks lack insulation from executive control, but becomes substantial where bureaucratic autonomy (15) to act independently from executive overreach is high (16). Institutional capacity to implement policies (17) aimed at managing environmental externalities independently from partisan pressures helps reduce emissions. Central bank independence and its limits are often defined by statutes (18). This process entails deliberations between members of the executive and legislative powers (19), which can ultimately end in court (20), making it challenging for political leaders to undermine such regulatory powers (16).¹ These findings suggest that participation in international regulatory networks, conditioned by domestic institutional capacity and bureaucratic autonomy, shapes how financial regulation can contribute to decarbonization.

¹But see Fernández-Albertos (19) for a discussion of the potential politicization of central banks.

Central Banks, Climate Change, and Economic Policy

Central banks and financial supervisors shape where capital goes. By issuing prudential rules and regulations with which banks and investment funds must comply, central banks govern which sectors of the real economy receive financing (8). This feature has been historically documented when nations prioritized industrial, carbon-intensive development (5, 7, 21). As climate risks have emerged as a recognized systemic financial risk (1, 22), central banks have increasingly been called upon to extend its regulatory powers toward decarbonization (3, 4, 23). The NGFS—established at the Paris “One Planet Summit” in 2017 and spearheaded by financial supervisors from France, China, England, Germany, Mexico, Norway, Singapore, and Sweden—has worked to build a coordinating infrastructure for climate-aligned financial supervision (10, 11).

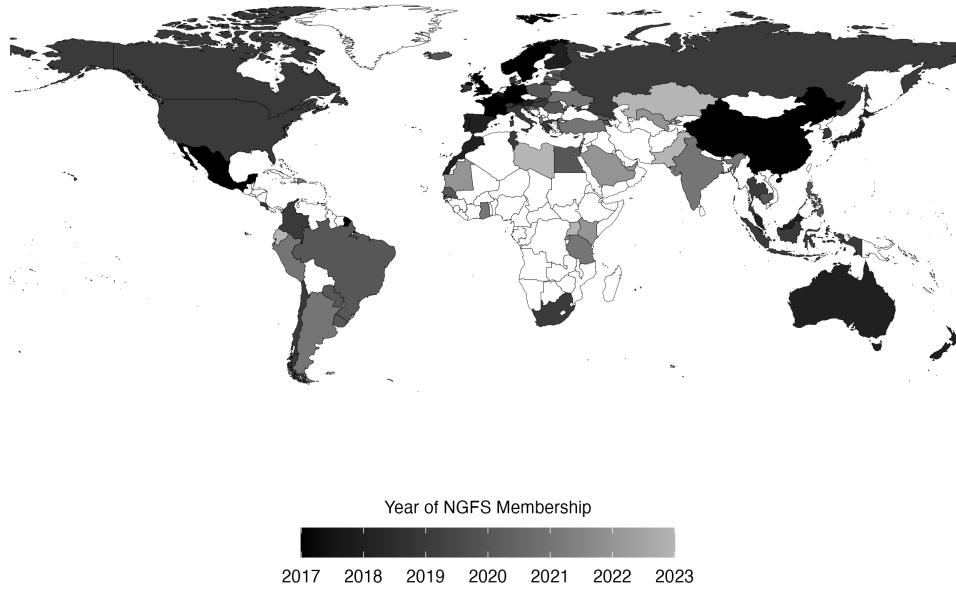
Financial regulators that join the NGFS commit to integrating climate scenarios into financial supervision and to standardizing the classification of “green” and “brown” assets (1, 11). One goal is to facilitate capital reallocation from carbon-intensive (i.e., “brown”) to low-carbon (i.e., “green”) ventures. Some NGFS members have translated these commitments into binding domestic regulations. For example, in 2021, Brazil’s central bank mandated that financial institutions disclose climate liabilities under their Social, Environmental, and Climate Responsibility Policy (PRSAC)² documents (16). The Brazilian central bank relied on its bureaucratic autonomy to implement policies without executive interference (16). By contrast, the U.S. Federal Reserve—constrained by political opposition—joined the NGFS only during the Biden administration (16). This contrast illustrates that central bank independence (14) is not merely a macroeconomic property but a precondition for effective climate governance.

A natural skeptical reaction to such international commitments is that NGFS membership might represent “performative governance” (24, 25). That is, in response to their citizens’ pressure, wealthy, industrialized nations that have already been on a decarbonization path may have joined the NGFS because the costs of membership were low and the reputational benefits were high. Based on this argument, any observed emissions gap between NGFS members and non-members would reflect pre-existing trajectories rather than an effect stemming from NGFS participation. This concern is compounded by the network’s initial geographic concentration in the Global North (see Fig. 1), where decarbonization pressures from domestic constituencies, regional regulators, and capital markets were already substantial before 2017. Importantly, as panel B of Fig. 1 shows, observed emission trajectories show that NGFS members were, in fact, larger emitters than non-members before 2017, with a modest, almost imperceptible downward trend among nations that joined the NGFS in 2017-2018. If wealthy, already-greening nations had self-selected into the network, a steeper downward trajectory would be expected instead.

²In Portuguese, PRSAC refers to *Política de Responsabilidade Social, Ambiental e Climática*.

A

World Map by NGFS Membership



B

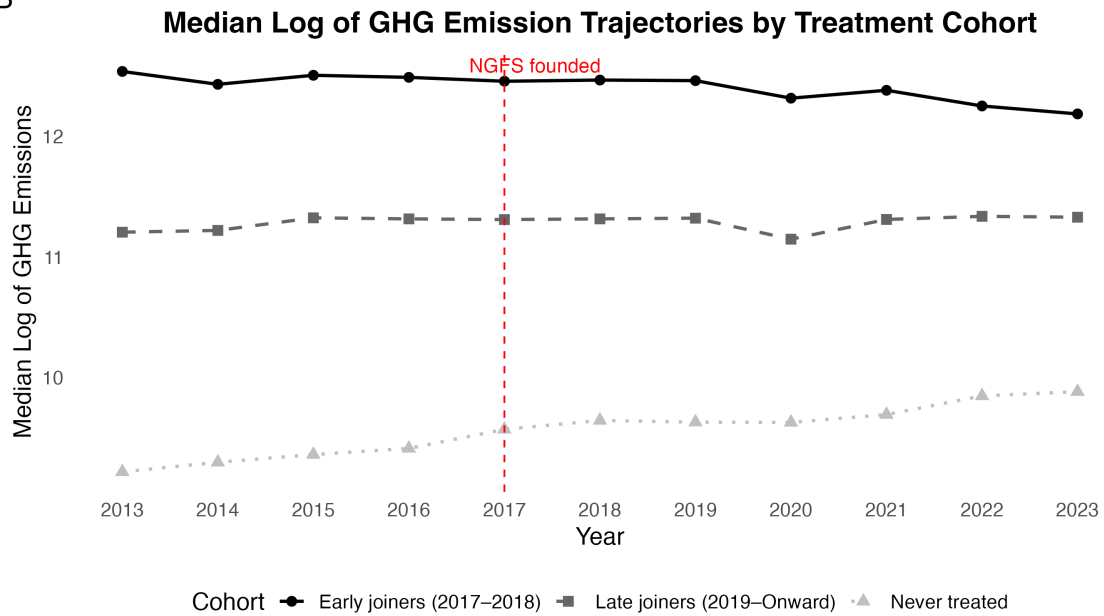


Fig. 1

A rigorous empirical test must nonetheless account for systematic differences between early joiners (2017–2018), late joiners (2018–onward), and never-treated countries. This step would address selection bias concerns and help demonstrate that any post-membership emissions

deceleration exceeds what pre-existing within-country trends would predict. I conduct such analysis using panel data from 145 countries between 2013 and 2023. As I demonstrate, NGFS membership is associated with lower within-country emissions. Additionally, central bank independence moderates the magnitude of the emissions effect that NGFS membership can realistically produce.

Results

NGFS membership is associated with lower within-country emissions across multiple specifications

To estimate the effect of joining the NGFS on national GHG emissions, I fit a series of two-way fixed effects (TWFE) models with country and year fixed effects. Country-level fixed effects absorb time-invariant differences across nations, e.g., natural resources characteristics and institutional developmental paths. Year-level fixed effects account for global shocks common to all countries simultaneously, including economic downturns, global pandemics, and multilateral climate agreements, such as the Paris Agreement. Accordingly, the identifying variation in the models is within-country: changes in emissions trajectories following NGFS entry, relative to never-treated countries in the same years.

I assess the effect across three outcomes: total GHG emissions, emissions normalized by GDP per capita, and emissions per capita. All models control for economic, ecological, and political variables, such as trade openness, land use, and institutional capacity. Fig. 2 shows that NGFS membership is associated with a statistically significant within-country reduction in emissions across all three specifications. Point estimates range from approximately 4% to 9% lower emissions among NGFS members compared to non-members, all else equal. Despite differences in the confidence intervals when emissions are normalized by population or GDP per capita, the direction and significance of the effect are consistent across specifications.

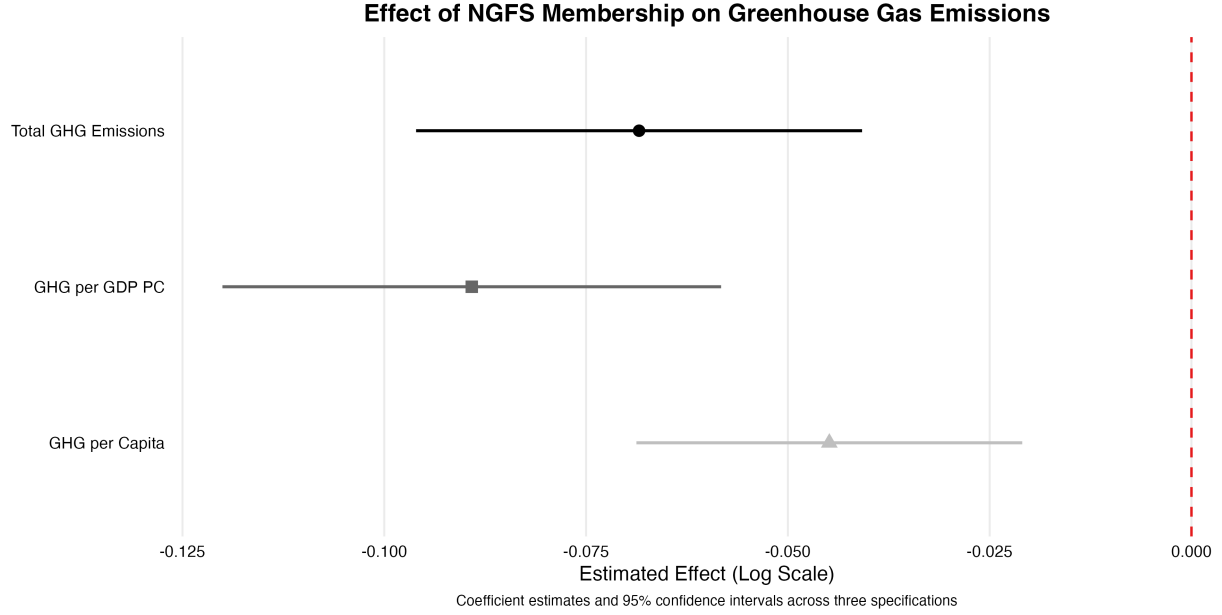


Fig. 2

Standard TWFE estimates can be vulnerable to bias under staggered adoption; a doubly robust event-study design addresses both timing heterogeneity and selection concerns

Yet, two main challenges complicate the TWFE results displayed in Fig. 2. First, countries joined the NGFS at different times—i.e., observations had a staggered adoption of the NGFS treatment. France co-founded the network in 2017, Indonesia joined in 2019, Jordan in 2021, and so on. Also, recent work shows that TWFE can produce biased estimates when treatment effects are heterogeneous across adoption cohorts (12, 26, 27). Second, early NGFS members are disproportionately wealthy, industrialized economies, raising the concern that any observed emissions gap reflects pre-existing divergence rather than network membership.

To address both issues simultaneously, I employ Callaway and Sant’Anna’s doubly robust difference-in-differences estimator and aggregate the resulting group-time average treatment effects using a dynamic aggregation. This technique restricts comparisons to never-treated countries and reweights observations via propensity scores to improve pre-treatment comparability across cohorts. Fig. 3 shows the CS dynamic estimates for various measures of emissions, revealing a post-treatment downward trajectory across all three specifications. The post-treatment decline is stronger and more precisely estimated for total GHG emissions and GHG per GDP per capita than for GHG per capita.

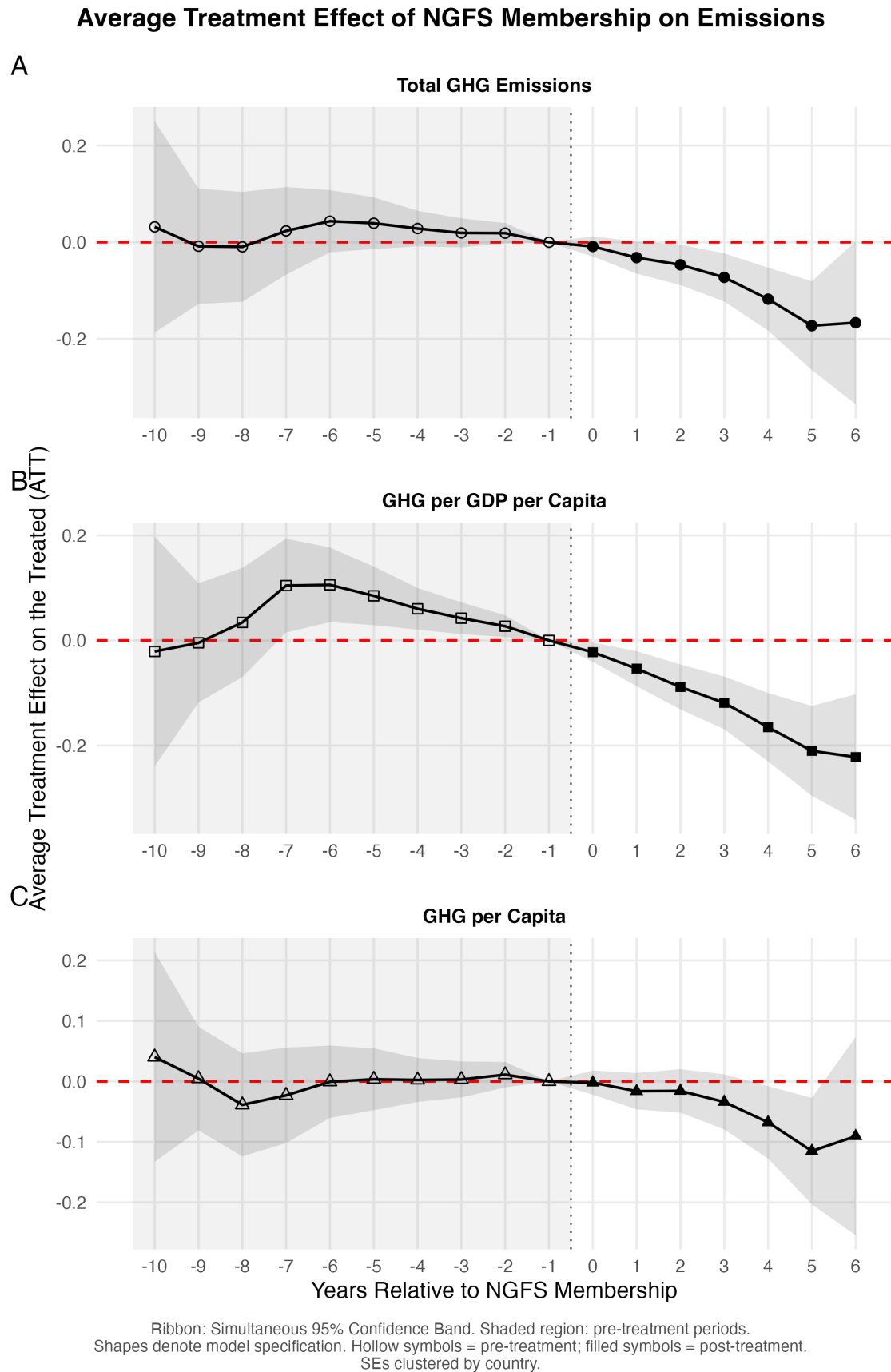


Fig. 3
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Pre-membership estimates, in turn, for emissions normalized per GDP per capita reveal distinct trajectories. The observable differences potentially stem from early-joiners being wealthier, more industrialized nations than late-joiners and never-treated countries (see Fig. 1). This point warrants an honest acknowledgement: Pre-tests of parallel trends reveal statistically significant effects particularly among early-joiners between 2013 and 2015, violating the parallel trends assumptions ($p \approx 0$). I further conducted a formal sensitivity analysis (26), which shows that the overall effect on GHG emissions is robust to modest violations of parallel trends ($\bar{M} \approx 0.13$). However, the effect does not resist violations comparable in size to the largest pre-treatment deviation (see Supplementary Materials).

At the same time, Fig. 1 details how the pre-treatment estimates follow a slight positive-to-zero decline in the two to three years before treatment. The estimates within three years of treatment are small and statistically indistinguishable from zero across all cohorts. This feature, combined with the results of the sensitivity analysis, provides relevant parallel trends evidence while considering the overall violation of the parallel trends concentrated in the early cohorts.

In sum, the findings are consistent with a modest pre-existing convergence between NGFS members and the never-treated control group. Also, the average treatment effect on the treated (ATT) group is significant and in the same direction as the TWFE results displayed in Fig. 2. The ATT for emissions reduction are found in the magnitude of 8% for overall emissions ($p < 0.05$), 13% for emissions per GDP per capita ($p < 0.1$), and 5% for emissions per capita ($p < 0.05$). Additionally, the observed data on cohort trajectories shown in Fig. 1 confirm that neither early-joiners nor late-joiners display a pronounced pre-existing downward trend in median emissions. Indeed, the doubly robust reweighting is designed to address the selection on observable pre-treatment characteristics. Therefore, the post-treatment effect identified in Fig. 3 should be interpreted as an accelerated deceleration in emission growth relative to trend, not a clean structural break from a perfectly flat baseline.

Central bank independence conditions the magnitude of the NGFS effect

The TWFE and DID results indicate that, all else equal, NGFS membership is associated with average lower within-country emissions. I argue that the capacity of central banks to translate international climate commitments into domestic regulatory action depends on their insulation from political interference (5, 7, 10, 16, 21). To test this argument, I interact NGFS membership with a Garriga’s (14) weighted central bank independence (CBI) index and estimate marginal effects across the observed range of CBI scores.

Fig. 4 reveals that the marginal effect of NGFS membership is a downward-sloping gradient across the CBI distribution, not a threshold or a discontinuity. At low central bank independence levels (one standard deviation below the global mean, $CBI \approx 0.42$), NGFS membership is associated with a 3.3% reduction in emissions ($\beta = -0.03$, $SE = 0.02$, $p < 0.05$). At the global mean ($CBI \approx 0.62$), the reduction increases to approximately 6.0% ($\beta = -0.06$, $SE = 0.01$, $p < 0.001$). Among countries with highly independent central banks ($CBI \approx 0.81$), the estimated NGFS membership effect reaches about 8.7% ($\beta = -0.09$, $SE = 0.02$, $p < 0.001$). As the confidence band crosses zero at the lower tail of the CBI distribution, this indicates that membership carries little statistically reliable impact where

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central banks are most constrained by executive power. Fig. 5 illustrates a similar pattern in predicted emission levels: the gap between NGFS members and non-members widens progressively as CBI rises, with statistically reliable divergence beginning at $CBI \approx 0.41$.

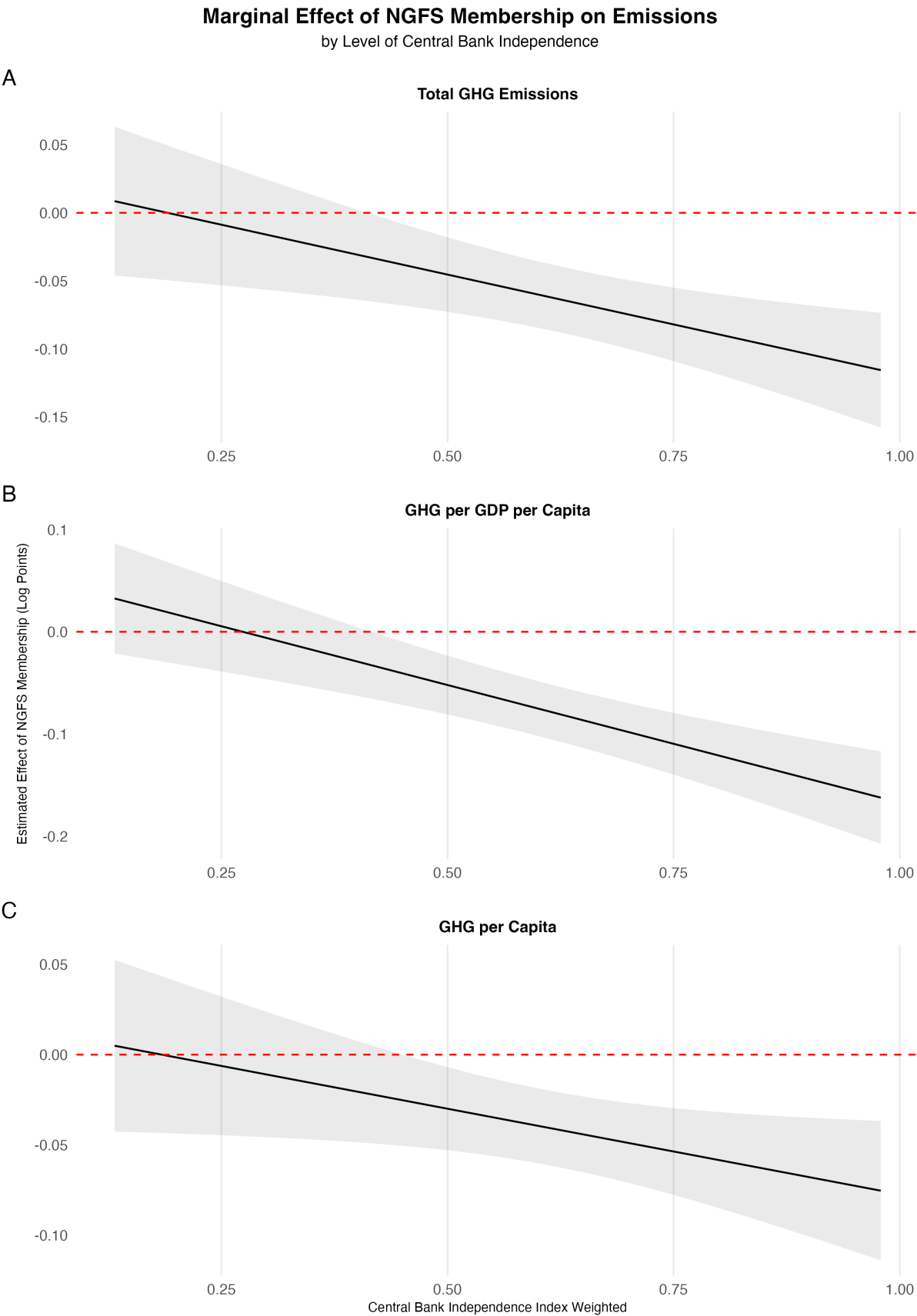
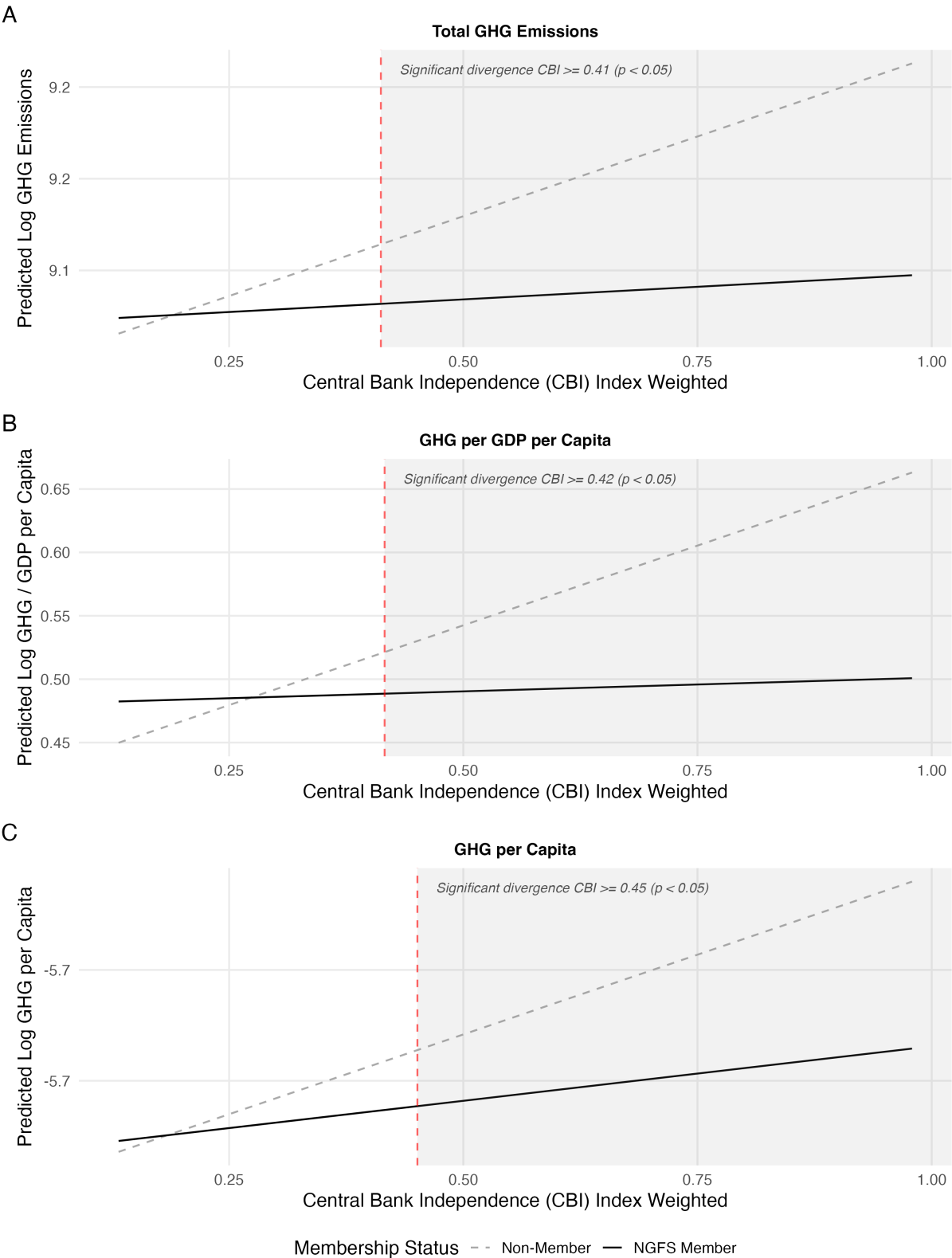


Fig. 4
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Predicted Emissions by NGFS Membership and CBI Level



Shaded region and dashed vertical line mark the CBI level above which the NGFS member vs. non-member divergence becomes statistically significant ($p < 0.05$); the threshold is model-specific.

Fig. 5

Central bank independence functions as a moderating condition, not as a mediating pathway activated by network membership

A plausible alternative mechanism is that joining the NGFS itself strengthens a central bank’s institutional autonomy. International network membership may provide legitimacy, peer pressure, and technical resources that insulate regulators from domestic political interference. This autonomy, in turn, may enhance and then produce the emissions reduction. To evaluate this mediation pathway, I conduct a causal mediation analysis using two-way demeaned within-country variation, with confidence intervals and p-values obtained from a cluster (block) bootstrap that resamples whole countries rather than individual country-years (see Supplementary Materials). The first-stage estimate reveals that NGFS membership is associated with a small but statistically significant *decrease* in CBI ($\beta = -0.01$, $p < 0.001$), not an increase. This pattern potentially reflects the coordination demands and institutional adjustments that network participation in the NGFS requires. The indirect effect through CBI is negligible in magnitude (proportion mediated $\approx 0.2\%$) and statistically indistinguishable from zero ($p = 0.82$), with its point estimate running counter to rather than reinforcing the overall reduction. The dominant pathway is the average direct effect (ADE) of NGFS membership ($ADE = -0.03$, $p < 0.001$). Taken together, these results indicate that pre-existing institutional autonomy amplifies the NGFS effect, but network membership does not generate autonomy as a byproduct. CBI is, therefore, a precondition for the network’s effectiveness in reducing within-country emissions, not a product of it.

Two features of this decomposition deserve a caveat. First, because CBI and state capacity are themselves plausibly related institutional features, each mediator model adjusts for the other candidate mediator: the CBI models control for state capacity, and the state-capacity models control for CBI. This choice isolates each pathway from the other, but it assumes the two mediators do not share a causal relationship with one another; if institutional capacity flows into central bank independence, or vice versa, as part of the same underlying process, mutually adjusting for both may partial out signal that a model treating them jointly would retain. Second, as a check on this assumption, I repeat the same decomposition using state capacity, rather than CBI, as the candidate mediator. NGFS membership is not significantly associated with within-country state capacity ($\beta = -0.01$, $p = 0.20$), and the resulting indirect effect is small and statistically indistinguishable from zero (proportion mediated $\approx 0.8\%$, $p = 0.58$), reinforcing the overall reduction just as the CBI pathway does. Neither candidate mediator, in other words, is meaningfully shifted by NGFS membership, which reinforces rather than merely accompanies the conclusion reached through the CBI pathway above: pre-existing institutional characteristics condition the network’s effectiveness rather than being manufactured by it.

Discussion

This article asks whether countries with central banks formally committed to climate-aligned financial regulation emit less greenhouse gases than those without such commitment. Using NGFS membership as a cross-nationally comparable treatment indicator, I find that, after controlling for economic, ecological, and political factors, within-country GHG emissions decelerate following network entry. The reduction effect on emissions ranges from 4% to

9% depending on how GHG is measured (fully-controlled specification, Fig. 2). The effect is consistent across three emissions outcomes and is robust to two identification checks that address serious identification threats: heterogeneous treatment timing and endogenous selection into the network. Three findings deserve special attention for what they reveal about the mechanisms through which financial regulation, at the global and national levels, contributes to decarbonization.

First, joining the NGFS in itself is significantly associated with reduced emissions, but network membership alone is modest in absolute terms. The observed data on cohort trajectories displayed in Fig. 1 show no substantial level shift in emissions following network entry. Statistically, the effect on lowering emissions is part of the within-country fixed-effects residual, reflecting a deceleration of emission growth rather than an abrupt reduction. This is consistent with how financial regulation operates. Central banks do not function as environmental protection agencies or a ministry of the environment. Financial supervisors do not mandate emission cuts directly, but by gradually reshaping investment incentives, disclosure requirements, and risk pricing in ways that compound over time. Still, the modest magnitude of the effect in a seven-year window analyzed here should not be mistaken for evidence that financial regulation is unimportant; it may instead reflect the early stage of NGFS institutionalization during the study period, which might keep producing positive externalities for late joiners to continue reducing their emissions moving forward.

Second, the moderation by central bank independence has direct policy implications. The results shown in Fig. 4 indicate that NGFS membership does not affect all nations uniformly. The positive ramifications stemming from NGFS membership are largely concentrated among countries whose central banks can translate international climate commitments into binding domestic regulatory action as freely from political interference as possible. This finding speaks to a broader literature on the conditions under which international institutional membership produces domestic policy change (CITE). The NGFS is a voluntary, non-binding network. Its influence operates through deliberations, agenda-setting, and other incentives rather than legal obligations produced by hard international law. Where central banks lack the autonomy to act independently from executive power, membership may remain closer to a “performative governance” scenario (24).

Third, the mediation results complicate a narrative that joining the NGFS itself builds institutional capacity to reduce emissions by connecting central banks to international peers and resources (1, 4, 10). Instead, the evidence here suggests that NGFS membership is associated with a slight *reduction* in measured CBI. This finding is consistent with the view that network participation requires coordination and deference to shared standards that may modestly constrain domestic, unilateral regulatory discretion. At the same time, the dominant pathway from NGFS membership to emissions is direct, i.e., through the regulatory and supervisory channels the network establishes, rather than through a feedback loop that first builds domestic autonomy.

These conclusions have noteworthy limitations. The statistical analysis ends in 2021, capturing at most four post-treatment years for the founding members and fewer for later joiners. The event-study estimates show that the emissions reduction continues to deepen through year three. This trend suggests that the full long-run effect may be larger than

what is estimable here. Extending the analysis period could yield more comprehensive insights and check whether the effect persists beyond the examined timeframe. My assessment also treats NGFS membership as a binary indicator, obscuring variation in the depth of implementation across members. For instance, Brazil devised a binding PRSAC mandate (16), whereas the U.S. Federal Reserve’s presented a more constrained engagement. Such contrast represents very different degrees of practical commitment that a single membership dummy variable cannot capture. Future research drawing from qualitative insights, case studies, and implementation-quality data could better identify causal mechanisms within the NGFS’s broad treatment indicator used here.

Even considering these caveats, the results have a clear implication: financial regulation is a meaningful instrument of climate policy, but its effectiveness is conditioned on the international and domestic institutional frameworks within which financial regulators operate. Efforts to expand the NGFS and similar networks may produce greater emissions reductions where they are paired with, or preceded by, investments in central bank institutional capacity and bureaucratic autonomy. For countries in the Global South, where both NGFS membership is still growing and central bank independence remains lower on average, the remaining questions about which conditions may further enable emissions reduction is not merely academic. It has practical consequences for how international climate finance governance should be designed for the global economy to move toward decarbonization as countries in the Global North and Global South alike pursue their net-zero goals.

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